

Calibration of GARCH(1,1) Model from Historical Prices

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First version: March 2, 2020

Revised: December 12, 2025

1 Overview

The Generalized Autoregressive Conditional Heteroskedasticity (GARCH) model, typically GARCH(1,1), is calibrated using Maximum Likelihood Estimation (MLE) on the asset's log returns (r_t).

2 Step 1: Data Transformation

Log Returns Calculation

The historical price series (P_t) must be converted into a time series of log returns (r_t):

$$r_t = \ln \left(\frac{P_t}{P_{t-1}} \right)$$

Stylized Facts

The returns are expected to exhibit **volatility clustering** and **heavy tails**.

3 Step 2: Model Specification (GARCH(1,1))

The model is defined by a Mean Equation and a Variance Equation.

A. Mean Equation (AR(0))

The simplest assumption is a constant mean μ for the returns:

$$r_t = \mu + \epsilon_t$$

where ϵ_t is the innovation or shock term.

B. Innovation Definition

The shock term ϵ_t is the product of the conditional volatility (σ_t) and a white noise process (z_t):

$$\epsilon_t = \sigma_t z_t$$

with $z_t \sim \text{i.i.d. } D(0,1)$. The distribution D is often assumed to be Normal ($\mathcal{N}$) or Student's t .

C. Variance Equation (GARCH(1,1))

The conditional variance σ_t^2 is modeled as a function of the long-run variance (ω), the previous shock squared (ϵ_{t-1}^2), and the previous conditional variance (σ_{t-1}^2):

$$\sigma_t^2 = \omega + \alpha \epsilon_{t-1}^2 + \beta \sigma_{t-1}^2$$

D. Parameter Constraints

The parameters must satisfy the following constraints for stationarity and positivity:

- Positivity: $\omega > 0, \alpha \geq 0, \beta \geq 0$
- Stationarity: $\alpha + \beta < 1$

4 Step 3: Parameter Estimation (MLE)

The parameters $\Theta = \{\mu, \omega, \alpha, \beta, (\nu)\}$ are estimated by maximizing the log-likelihood function ($\mathcal{L}$).

A. Log-Likelihood Function (Assuming Normal Distribution)

The likelihood for a sample of T returns, assuming $z_t \sim \mathcal{N}(0, 1)$, is:

$$\mathcal{L}(\Theta|r_1, \dots, r_T) = \sum_{t=1}^T \ell_t(\Theta)$$

where the log-likelihood contribution at time t is:

$$\ell_t(\Theta) = -\frac{1}{2} \left[\ln(2\pi) + \ln(\sigma_t^2) + \frac{(r_t - \mu)^2}{\sigma_t^2} \right]$$

The objective is to find $\hat{\Theta} = \arg \max_{\Theta} \mathcal{L}(\Theta)$.

B. Numerical Optimization

Optimization algorithms (e.g., quasi-Newton methods) are used to iteratively find the maximum likelihood estimates $\hat{\Theta}$ since the function is non-linear.

5 Step 4: Diagnostics and Validation

After estimation, the model must be validated using the standardized residuals $\hat{z}_t$:

$$\hat{z}_t = \frac{r_t - \hat{\mu}}{\hat{\sigma}_t}$$

1. **No Autocorrelation in Residuals:** $\hat{z}_t$ must be uncorrelated (tested via Ljung-Box test).
2. **No Remaining ARCH Effects:** $\hat{z}_t^2$ must be uncorrelated (tested via Ljung-Box or ARCH-LM test). If correlation exists, the GARCH specification (e.g., GARCH(1,1)) is insufficient.
3. **Significance:** Estimated parameters $(\hat{\omega}, \hat{\alpha}, \hat{\beta})$ should be statistically significant (based on p-values).

6 Step-by-Step GARCH Calibration Algorithm

The objective is to estimate the parameters $\Theta = \{\mu, \omega, \alpha, \beta\}$ by maximizing the log-likelihood function $\mathcal{L}(\Theta)$.

7 Illustrative Example with 5 Historical Prices

We illustrate Steps 1, 3, and the recursive part of Step 4 using a minimal dataset.

Table 1: GARCH Calibration Algorithm

Step	Description	Formula / Condition
1. Data Prep	Convert the price series (P_t) to log returns (r_t).	$r_t = \ln\left(\frac{P_t}{P_{t-1}}\right)$
2. Model Spec	Define the GARCH(1,1) system equations.	$\sigma_t^2 = \omega + \alpha\epsilon_{t-1}^2 + \beta\sigma_{t-1}^2$
3. Initialize	Set initial values for parameters $\Theta^{(0)}$ and the starting variance (σ_0^2).	$\sigma_0^2 = \sigma_{LR}^2 = \frac{\omega}{1-\alpha-\beta}$
4. Likelihood	Recursively calculate σ_t^2 , innovation ϵ_t , and log-likelihood contribution ℓ_t for $t = 1$ to T .	$\ell_t = -\frac{1}{2} \left[\ln(\sigma_t^2) + \frac{(r_t - \mu)^2}{\sigma_t^2} \right]$
5. Optimize	Use a numerical method (e.g., BFGS) to find the parameter set $\hat{\Theta}$ that maximizes $\mathcal{L}(\Theta) = \sum_{t=1}^T \ell_t$.	$\hat{\Theta} = \arg \max_{\Theta} \mathcal{L}(\Theta)$
6. Diagnose	Validate the model by checking standardized residuals ($\hat{z}_t$) for autocorrelation and ensuring constraints are met.	Constraints: $\omega, \alpha, \beta \geq 0, \alpha + \beta < 1$

Data Input and Log Returns (r_t)

Time (t)	Price (P_t)	Log Return (r_t)
0	100.00	N/A
1	101.50	$r_1 \approx 0.014888$
2	100.50	$r_2 \approx -0.009901$
3	99.00	$r_3 \approx -0.015038$
4	102.00	$r_4 \approx 0.029945$

Initial Parameters and Calculation (Step 4 Iterations)

Initial Guesses $\Theta^{(0)}$: $\mu = 0, \omega = 0.00001, \alpha = 0.05, \beta = 0.90$.

Initial Variance: $\sigma_0^2 = 0.000200$.

Table 2: Recursive Conditional Variance Calculation

Time (t)	r_t	ϵ_{t-1}^2	$\sigma_t^2 = \omega + \alpha\epsilon_{t-1}^2 + \beta\sigma_{t-1}^2$	ϵ_t^2
0	N/A	N/A	$\sigma_0^2 = 0.000200$ (Init)	N/A
1	0.014888	$\sigma_0^2 = 0.000200$	$\sigma_1^2 \approx 0.00020000$	0.00022165
2	-0.009901	0.00022165	$\sigma_2^2 \approx 0.00020008$	0.00009803
3	-0.015038	0.00009803	$\sigma_3^2 \approx 0.00019507$	0.00022614
4	0.029945	0.00022614	$\sigma_4^2 \approx 0.00018784$	0.00089670

The optimization (Step 5) would then use these calculated values to find the total log-likelihood $\mathcal{L}(\Theta^{(0)})$ and refine the parameter estimates.